Using AI to improve your code

Eris :3 (they/them)

October 23, 2026

I FUCKING HATE AI!!!

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These are my thoughts on ai :) Artificial intelligence will not be mentioned again for the rest of this talk.

Using ~~AI~~ Abstract Interpretation to improve your code

Eris :3 (they/them)

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How other people use Abstract Interpretation to improve your code when you run it

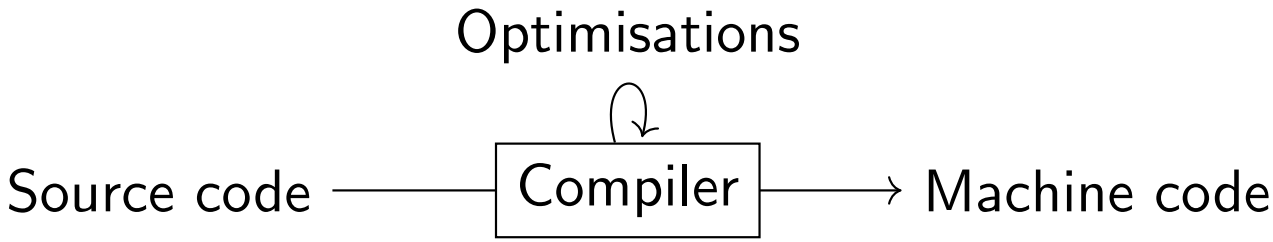
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Optimising compilers



Optimising compilers



Fibonacci program

$F \leftarrow \underbrace{[0, \ldots, 0]}_{\text{100 of them}}$
$F[0] \leftarrow 0$
$F[1] \leftarrow 1$
$i \leftarrow 2$
while $i < 100$ **do**
    $F[i] \leftarrow F[i-1] + F[i-2]$
    $i \leftarrow i + 1$
end while
return F

Fibonacci program

$F \leftarrow \underbrace{[0, \ldots, 0]}_{\text{100 of them}}$
if $\neg(0 \leq 0 \leq 99 \wedge 0 \leq 1 \leq 99)$ **then**
    explode (out of bounds)
end if
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if false **then**
    explode (out of bounds)
end if
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$i \leftarrow 2$
while $i < 100$ **do**
    if $\neg(0 \leq i-2 \leq 99 \wedge 0 \leq i-1 \leq 99 \wedge 0 \leq i \leq 99)$ **then**
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    $i \leftarrow i + 1$
end while
return F

The bounds check is redundant!

Syntax

Expressions, conditions and statements are defined inductively. Let n be numeric literals and v variables.

$$
\begin{array}{lcl}
e & := & n \mid v \mid e + e \mid e - e \mid e \times e \\
c & := & \mathrm{true} \mid \mathrm{false} \mid e \leq e \mid e = e \mid c \wedge c \mid c \vee c \mid \neg c \\
s & := & v \leftarrow e \mid s; s \mid \textbf{if } c \textbf{ then } s \textbf{ else } s \textbf{ end if} \\
 & & \mid \textbf{while } c \textbf{ do } s \textbf{ end while} \mid \textbf{explode} \mid \textbf{skip}
\end{array}
$$

Oh wait arrays

Let a be array variables. Then we can extend expressions and statements to include array syntax.

$$\begin{array}{rcl} e & := & \cdots \mid a[e] \\ s & := & \cdots \mid a[e] \leftarrow e \end{array}$$

Semantics

There is a set $\mathcal{E}$ of states (E for environment i think?) and (letting $\mathcal{D} = \mathcal{P}(\mathcal{E})$) an operator $\mathbb{S}[\![s]\!] : \mathcal{D} \to \mathcal{D}$ for each statement s. This operator describes "what the statements do".

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Let's write it out! Let $\mathcal{E} = \mathbb{Z}^V \times A^W$ where V is the set of variables and W is the set of array variables.

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$$A = \bigcup_{n \in \mathbb{N}} \{n\} \times \mathbb{Z}^n$$

so like each array has a number telling us its length and then a whole bunch of integers...

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$$A = \bigcup_{n \in \mathbb{N}} \{n\} \times \mathbb{Z}^n$$

so like each array has a number telling us its length and then a whole bunch of integers... and then we can define the semantics operator..... that sounds SOOOO annoying!! these arrays are so ugly! i don't wanna

Order theory 1

Definition

A poset is a set P equipped with a partial order $\sqsubseteq$ (reflexive, antisymmetric and transitive).

(I really hope you've seen this before else this talk will go way too fast)

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Example

For any set S, $(\mathcal{P}(S), \subseteq)$ is a poset.

Order theory 1

Insert hasse diagram here

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Definition

A map $f : P \to Q$ between posets is monotone if whenever $x \sqsubseteq_P y$, $f(x) \sqsubseteq_Q f(y)$.

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Proposition

The identity map is monotone. Composites of monotone maps are monotone.

Proof.

If $x \sqsubseteq y$ then $x \sqsubseteq y$.
If $x \sqsubseteq y$ then $f(x) \sqsubseteq f(y)$ then $g(f(x)) \sqsubseteq g(f(y))$. □

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Monotone maps are ordered pointwise, $f \sqsubseteq g$ iff $f(x) \sqsubseteq g(x)$ for all x in the domain.

Approximating semantics

Instead of defining the full semantics, define an approximation set of states $\hat{\mathcal{E}}$ and (letting $\hat{\mathcal{D}} = \mathcal{P}(\hat{\mathcal{E}})$) define an approximation operator $\hat{\mathbb{S}}[\![s]\!] : \hat{\mathcal{D}} \to \hat{\mathcal{D}}$ for each statement s.

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Need a monotone map $\hat{\gamma} : \hat{\mathcal{D}} \to \mathcal{D}$ such that $\mathbb{S}[\![s]\!] \circ \gamma \sqsubseteq \gamma \circ \hat{\mathbb{S}}[\![s]\!]$.

$$\begin{array}{ccc} \hat{\mathcal{D}} & \xrightarrow{\hat{\mathbb{S}}[\![s]\!]} & \hat{\mathcal{D}} \\ {\scriptstyle\gamma}\downarrow & \swarrow\!\!\!\swarrow & \downarrow{\scriptstyle\gamma} \\ \mathcal{D} & \xrightarrow[\mathbb{S}[\![s]\!]]{} & \mathcal{D} \end{array}$$

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If you prefer, we need $\hat{\gamma}$ a lax natural transformation viewing semantics opeators as functors from the one object category of statements under sequential composition to the poset-enriched category of posets!

Approximating semantics

Define $\hat{\mathcal{E}} = \mathbb{Z}^V$ (no arrays!) and $\hat{\mathcal{D}} = \mathcal{P}(\hat{\mathcal{E}})$. Then $\gamma : \hat{\mathcal{D}} \to \mathcal{D}$ by $\gamma(S) = S \times A^W$, giving all possible assignments of array variables, is an approximation.
Then we'll define $\mathbb{E}[\![e]\!] : \hat{\mathcal{E}} \to \mathcal{P}(\mathbb{Z})$, $\mathbb{C}[\![c]\!] : \hat{\mathcal{D}} \to \hat{\mathcal{D}}$ and $\hat{\mathbb{S}}[\![s]\!] : \hat{\mathcal{D}} \to \hat{\mathcal{D}}$ inductively.

Approximating semantics

$$
\begin{aligned}
\mathbb{E}[\![n]\!](\sigma) &= \{n\} \\
\mathbb{E}[\![v]\!](\sigma) &= \{\sigma(v)\} \\
\mathbb{E}[\![e_1 + e_2]\!](\sigma) &= \{x + y \mid x \in \mathbb{E}[\![e_1]\!](\sigma), y \in \mathbb{E}[\![e_2]\!](\sigma)\} \\
\mathbb{E}[\![e_1 - e_2]\!](\sigma) &= \{x - y \mid x \in \mathbb{E}[\![e_1]\!](\sigma), y \in \mathbb{E}[\![e_2]\!](\sigma)\} \\
\mathbb{E}[\![e_1 \times e_2]\!](\sigma) &= \{x \times y \mid x \in \mathbb{E}[\![e_1]\!](\sigma), y \in \mathbb{E}[\![e_2]\!](\sigma)\} \\
\mathbb{E}[\![a[i]]\!](\sigma) &= \mathbb{Z}
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{C}[\![\text{true}]\!](S) &= S \\
\mathbb{C}[\![\text{false}]\!](S) &= \emptyset \\
\mathbb{C}[\![e_1 \leq e_2]\!](S) &= \{\sigma \in S \mid \exists x \in \mathbb{E}[\![e_1]\!], y \in \mathbb{E}[\![e_2]\!] : x \leq y\} \\
\mathbb{C}[\![e_1 = e_2]\!](S) &= \{\sigma \in S \mid \exists x \in \mathbb{E}[\![e_1]\!], y \in \mathbb{E}[\![e_2]\!] : x = y\} \\
\mathbb{C}[\![c_1 \wedge c_2]\!](S) &= \mathbb{C}[\![c_1]\!](S) \cap \mathbb{C}[\![c_2]\!](S) \\
\mathbb{C}[\![c_1 \vee c_2]\!](S) &= \mathbb{C}[\![c_1]\!](S) \cup \mathbb{C}[\![c_2]\!](S) \\
\mathbb{C}[\![\neg c]\!](S) &= \hat{\mathcal{D}} - \mathbb{C}[\![c]\!](S)
\end{aligned}
$$

Approximating semantics

$$\hat{\mathbb{S}}[\![s_1 ; s_2]\!](S) = \hat{\mathbb{S}}[\![s_2]\!](\hat{\mathbb{S}}[\![s_1]\!](S))$$
$$\hat{\mathbb{S}}[\![v \leftarrow e]\!](S) = \{\sigma[v \mapsto \mathbb{E}[\![e]\!](\sigma)] \mid \sigma \in S\}$$
$$\hat{\mathbb{S}}[\![a[i] \leftarrow e]\!](S) = S$$
$$\hat{\mathbb{S}}[\![\textbf{if } c \textbf{ then } s_1 \textbf{ else } s_2 \textbf{ end if}]\!](S) = \hat{\mathbb{S}}[\![s_1]\!](\mathbb{C}[\![c]\!](S)) \cup \hat{\mathbb{S}}[\![s_1]\!](\mathbb{C}[\![\neg c]\!](S))$$

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We need some sort of limit of repeatedly applying s! Alternatively, we could find some minimal X such that $X = X \cup \hat{\mathbb{S}}[\![s]\!](\mathbb{C}[\![c]\!](X))$.

Order theory 2

Definition

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An ω-chain in a poset $(P, \sqsubseteq)$ is the image of a monotone map $f : \omega \to P$. Equivalently, it is a sequence $(x_i)_{i \in \omega}$ such that if $i \sqsubseteq j$ then $x_i \sqsubseteq x_j$.

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Definition

An ω complete partial order, or ω-CPO is a poset $(C, \sqsubseteq)$ such that all ω-chains (x_i) admit a least upper bound $\bigsqcup_i x_i$ (so $x_j \sqsubseteq \bigsqcup_i x_i$ for all j and is minimal among such). We may call these least upper bounds *joins*. An ω-CPO is *pointed* if it has a least element $\bot$.

Order theory 2

Insert hasse diagram here

Order theory 2

Example

For any set X, the poset $(\mathcal{P}(X), \subseteq)$ is an ω-cpo! Joins are given by unions of sets, $\bigsqcup_i X_i = \bigcup_i X_i$. Moreover, it is pointed by $\emptyset$!

We can think of joins as generalisations of unions, but we will only see instances of non-union joins later.

Order theory 2

Definition

A monotone map $f : C \to D$ of ω-cpos is ω-continuous if for any ω-chain $(x_i)_{i \in \omega}$ in C, $f(\bigsqcup_i x_i) = \bigsqcup_i f(x_i)$.

Theorem (Kleene fixed-point theorem for ω-cpos)

If C is a pointed ω-cpo and $f : C \to C$ is an ω-continuous map, then a fixed point of f exists, and the least fixed point is

$$\mathrm{lfp}(f) = \bigsqcup_i f^i(\bot).$$

Order theory 2

Proof.

Notice that f^i is monotone since it is a composite of monotone maps. Applied to $\bot \sqsubseteq f(\bot)$, we get $f^i(\bot) \sqsubseteq f^{i+1}(\bot)$, so $f^i(\bot)$ does form an ω-chain!



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Next, our computation does give a fixed point, since

$$f\left(\bigsqcup_i f^i(\bot)\right) = \bigsqcup_i f^{i+1}(\bot) = \bigsqcup_i f^i(\bot).$$

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Next, our computation does give a fixed point, since

$$f\left(\bigsqcup_i f^i(\bot)\right) = \bigsqcup_i f^{i+1}(\bot) = \bigsqcup_i f^i(\bot).$$

Finally, if x is another fixed-point of f, then applying monotonicity to $\bot \sqsubseteq x$ we get $f^i(\bot) \sqsubseteq f^i(x) = x$, so $\sqsubseteq_i f^i(\bot) \sqsubseteq x$. $\square$

While loops

And now we can define while loop semantics!

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$$\hat{\mathbb{S}}[\![\textbf{while } c \textbf{ do } s \textbf{ end while}]\!](S) = \mathbb{C}[\![\neg c]\!](\mathrm{lfp}(\lambda X.X \cup \hat{\mathbb{S}}[\![s]\!](\mathbb{C}[\![c]\!](X))))$$

We've finally finished defining the language ...

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We've finally finished defining the language ...
(There are questions about how we can know the inner expression is continuous if we haven't fully defined it but shhh this talk isn't about induction)

Abstract interpretation

The semantics we just described is very non-computable! Abstract interpretation is about approximating our "concrete" semantics so that it becomes computable.

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That is, we want a poset $(A, \sqsubseteq)$ with elements of A easy to represent on a computer, and some computable semantics operator $\mathbb{S}^\sharp[\![s]\!] : A \to A$ for each statement s, that's an approximation of our semantics by a map $\gamma : A \to \hat{\mathcal{D}}$.

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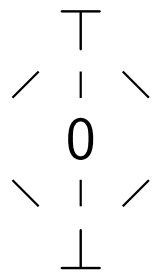
One approach is to store the sets of all values a variable could have ($\mathcal{D}' = \mathcal{P}(\mathbb{Z})^V$). We can then approximate $\mathcal{P}(\mathbb{Z})$ by A, and then get some computable set A^V!

Value domains

Examples of approximations of $\mathcal{P}(\mathbb{Z})$:

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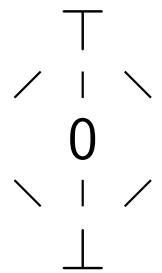
- $S =$ 

.

  $\gamma(+) = \{n \in \mathbb{Z} \mid n > 0\}$ etc.

Value domains

Examples of approximations of $\mathcal{P}(\mathbb{Z})$:

- $S =$ 

$\gamma(+) = \{n \in \mathbb{Z} \mid n > 0\}$ etc.
- $A = \{[a, b] \mid a, b \in \mathbb{Z} \cup \{-\infty, \infty\}; a \leq b\}$ ordered by subsets.
$\gamma([a, b]) = \{n \in \mathbb{Z} \mid a \leq n \leq b\}$.

Example again

$F \leftarrow \underbrace{[0, \ldots, 0]}_{\text{100 of them}}$
$F[0] \leftarrow 0$
$F[1] \leftarrow 1$
$i \leftarrow 2$
while $i < 100$ **do**
    if $2 > i \vee i > 99$ **then**
        explode (out of bounds)
    end if
    $F[i] \leftarrow F[i-1] + F[i-2]$
    $i \leftarrow i + 1$
end while
return F

At the second bounds check, $i \in [2, 99]$!!

Value set semantics

We define $\mathbb{E}'[\![e]\!] : \mathcal{P}(\mathbb{Z})^V \to \mathcal{P}(\mathbb{Z})$, $\mathbb{C}'[\![c]\!] : \mathcal{P}(\mathbb{Z})^V \to \mathcal{P}(\mathbb{Z})^V$ and $\mathbb{S}'[\![s]\!] : \mathcal{P}(\mathbb{Z})^V \to \mathcal{P}(\mathbb{Z})^V$.

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$$
\begin{aligned}
\mathbb{E}'[\![n]\!](\sigma) &= \{n\} \\
\mathbb{E}'[\![v]\!](\sigma) &= \sigma(v) \\
\mathbb{E}'[\![e_1 + e_2]\!](\sigma) &= \{x + y \mid x \in \mathbb{E}'[\![e_1]\!](\sigma), y \in \mathbb{E}'[\![e_2]\!](\sigma)\} \\
\mathbb{E}'[\![e_1 - e_2]\!](\sigma) &= \{x - y \mid x \in \mathbb{E}'[\![e_1]\!](\sigma), y \in \mathbb{E}'[\![e_2]\!](\sigma)\} \\
\mathbb{E}'[\![e_1 \times e_2]\!](\sigma) &= \{x \times y \mid x \in \mathbb{E}'[\![e_1]\!](\sigma), y \in \mathbb{E}'[\![e_2]\!](\sigma)\} \\
\mathbb{E}'[\![a[i]]\!](\sigma) &= \mathbb{Z}
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{C}'[\![\text{true}]\!](\sigma) &= \sigma \\
\mathbb{C}'[\![\text{false}]\!](\sigma) &= \bot \\
\mathbb{C}'[\![v \leq e]\!](\sigma) &= \sigma[v \mapsto \{n \in \sigma(v) \mid n \leq \mathbb{E}'[\![e]\!](\sigma)] \\
\mathbb{C}'[\![e_1 = e_2]\!](\sigma) &= \sigma \\
\mathbb{C}'[\![\neg c]\!](\sigma) &= \sigma \\
&\text{etc}
\end{aligned}
$$

Value set semantics

$$\mathbb{S}'[\![v \leftarrow e]\!](S) = \sigma[v \mapsto \mathbb{E}'[\![e]\!](\sigma)]$$
$$\mathbb{S}'[\![s_1 ; s_2]\!](S) = \mathbb{S}'[\![s_2]\!](\mathbb{S}'[\![s_1]\!](S))$$
$$\mathbb{S}'[\![a[i] \leftarrow e]\!](S) = S$$
$$\mathbb{S}'[\![\textbf{if } c \textbf{ then } s_1 \textbf{ else } s_2 \textbf{ end if}]\!](S) =$$
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From here, there is a trick to move to I^V from $\mathcal{P}(\mathbb{Z})^V$!

Order theory 3

oops this is out of place but needs to go somewhere

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Definition (Non-binary double feature AND lattice double feature?!)

A poset $(P, \sqsubseteq)$ is a join semi-lattice if every pair of elements x, y has a least upper bound (join) written $x \sqcup y$.

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Example

As usual $(\mathcal{P}(X), \sqsubseteq)$ is a join semi-lattice, where joins are unions!

Order theory 3

Definition

If $(C, \sqsubseteq)$ and $(A, \sqsubseteq)$ are posets, and $\alpha : C \rightleftarrows A : \gamma$ are monotone maps, then they form a Galois connection if $\alpha(c) \sqsubseteq a$ iff $c \sqsubseteq \gamma(a)$. We call α the left adjoint and γ the right adjoint. If γ is injective, then we call the pair of maps a Galois embedding.

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The injective map $\gamma([a, b]) = \{n \in \mathbb{Z} \mid a \leq n \leq b\}$ has left adjoint $\alpha(S) = [\min(S), \max(S)]$ forming a Galois embedding!

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The injective map $\gamma([a, b]) = \{n \in \mathbb{Z} \mid a \leq n \leq b\}$ has left adjoint $\alpha(S) = [\min(S), \max(S)]$ forming a Galois embedding!

Proposition

In a Galois embedding $\alpha : C \rightleftarrows A : \gamma$, $\alpha(\gamma(a)) = a$ for all $a \in A$.

Proof.

Firstly since $\gamma(a) \sqsubseteq \gamma(a)$, $\alpha(\gamma(a)) \sqsubseteq a$ so $\gamma(\alpha(\gamma(a))) \sqsubseteq \gamma(a)$. Next, since $\alpha(\gamma(a)) \sqsubseteq \alpha(\gamma(a))$, $\gamma(a) \sqsubseteq \gamma(\alpha(\gamma(a)))$. Hence $\gamma(\alpha(\gamma(a))) = \gamma(a)$ so by injectivity $\alpha(\gamma(a)) = a$. □

Order theory 3

Galois embeddings let us optimally approximate concrete structure in the abstract structure.

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Proposition

In a Galois embedding $\alpha : C \rightleftarrows A : \gamma$, if C is a join semi-lattice, then A is a join semi-lattice with join $a \sqcup b = \alpha(\gamma(a) \sqcup \gamma(b))$.

Proposition

Any operator $f : C \to C$ has its "best abstraction" on A (won't be defined sorry) by $\alpha \circ f \circ \gamma$.

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Any operator $f : C \to C$ has its "best abstraction" on A (won't be defined sorry) by $\alpha \circ f \circ \gamma$.

(I'm too dead to write the proofs for this... do them as exercises!) As a general idea, apply γ, then some operation, then α. This gives us approximation operators for free yay!

Computing Interval semantics

Let's write out what $\mathbb{E}^\sharp[\![e_1 + e_2]\!] : I^V \to I$ is.

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$$\begin{aligned}
\mathbb{E}^\sharp[\![e_1 + e_2]\!](\sigma^\sharp) &= \alpha(\mathbb{E}'[\![e_1 + e_2]\!](\gamma(\sigma^\sharp))) \\
&= \alpha(\{x + y \mid x \in [a, b], y \in [c, d]\}) \\
&= [\min(\{x + y \mid x \in [a, b], y \in [c, d]\}), \max(\cdots)] \\
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wow it's just interval arithmetic!

Remaining interval semantics

We still have to write out $\mathbb{C}^\sharp[\![c]\!]$ and $\mathbb{S}^\sharp[\![s]\!]$ ourselves.

$$\begin{aligned}
\mathbb{C}^\sharp[\![v < n]\!](\sigma^\sharp) &= \sigma[v \mapsto \sigma(v) \sqcap [-\infty, n-1]] \\
\mathbb{C}^\sharp[\![v = n]\!](\sigma^\sharp) &= \sigma[v \mapsto \sigma(v) \sqcap [n, n]] \text{ etc} \\
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Wow, this is almost all computable now!! The only thing that isn't is the while loop...

Fixpoint acceleration

Definition

An operation $\nabla : A \times A \to A$ on a join semi-lattice A is called a widening operator if $x \sqcup y \sqsubseteq x \nabla y$ and for any sequence $(y_i)_(i \in \omega)$, the sequence $x_0 = y_0$ and $x_{i+1} = x_i \nabla y_{i+1}$ is eventually constant.

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Theorem

If $g : A \to A$ is an approximation of $f : C \to C$ (in the sense of the (lax) commutative diagram from earlier), then the sequence $x_0 = \bot$ and $x_{i+1} = x_i \nabla g(x_i)$ stabilises in finite time to x to approximate the least fixed point of f, $\mathrm{lfp}(f) \sqsubseteq \gamma(x)$.

no proof...

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We can widen intervals like so:

$$[a, b] \nabla [c, d] = \left[\begin{cases} a & \text{if } a \leq c \\ -\infty & \text{otherwise} \end{cases}, \begin{cases} b & \text{if } b \geq d \\ \infty & \text{otherwise} \end{cases} \right].$$

Intuitively, if we see $[c, d]$ is growing, clamp to infinity!

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We can then define widening on I^V pointwise, and define while semantics with

$$\mathbb{S}^\sharp[\![\textbf{while } c \textbf{ do } s \textbf{ end while}]\!](S) = \mathbb{C}^\sharp[\![\neg c]\!](\mathrm{lfp}(\lambda X. X \nabla (S \sqcup \mathbb{S}^\sharp[\![s]\!](\mathbb{C}^\sharp[\![c]\!](X))))).$$

This is computable!

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This is computable!
Also everything here is an approximation of the concrete semantics i promise

Abstract interpreting

$F \leftarrow \underbrace{[0, \ldots, 0]}_{\text{100 of them}}$ $i \mapsto [\infty, \infty]$

$F[0] \leftarrow 0$

$F[1] \leftarrow 1$

$i \leftarrow 2$

while $i < 100$ **do**

  if $2 > i \vee i > 99$ **then**

    explode (out of bounds)

  end if

  $F[i] \leftarrow F[i-1] + F[i-2]$

  $i \leftarrow i + 1$

end while

return F

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we did it!!

Further reading



Patrick Cousot and Radhia Cousot.
Abstract interpretation: a unified lattice model for static analysis of programs by construction or approximation of fixpoints.
In Proceedings of the 4th ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages, POPL '77, page 238–252, New York, NY, USA, 1977. Association for Computing Machinery.



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